

1. REST API (Representational State Transfer)

What is it?

REST is an **architectural style** for APIs that is based on **resources**. Each entity (like user, product, etc.) is represented as a **URL**, and standard HTTP methods (`GET` , `POST` , `PUT` , `DELETE`) are used to interact with them.

Key Features:

- Access to **specific resources**
- Typically returns data in **JSON** format
- Requires **multiple calls** to fetch related data
- Simple and widely adopted

REST Example:

http

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`GET /users/123` → returns user 123's data

`GET /users/123/posts` → returns posts by user 123

2. Graph API (e.g., GraphQL)

What is it?

A **Graph API** (especially in the context of **GraphQL**) allows you to make **structured queries** that navigate relationships between entities. You can fetch **multiple related resources in a single request**.

Key Features:

- The client defines **exactly which data it needs**
- Can query **nested relationships**
- Ideal for complex, relational data (like social networks)
- Fewer requests → more efficient for composite queries

GraphQL Example:

graphql

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```
{
  user(id: 123) {
    name
    posts {
      title
      comments {
        content
      }
    }
  }
}
```

A single request returns the user, their posts, and the comments on those posts.



Key Differences

Feature	REST API	Graph API (GraphQL)
Structure	Resource-based (one URL per entity)	Graph-based (relationships between data)
Flexibility	Fixed response based on endpoint	Client defines exactly what to fetch
Number of calls	Often requires multiple	Typically just one
Over-fetching data	Common (may return more than needed)	Reduced (fetch only what you ask for)
Learning curve	Easier at first	Slightly steeper



When to Use Each?

- **Use REST API when:**
 - You have a simple data structure
 - You need quick and easy implementation
 - You're working with standardized services (e.g., public APIs)
- **Use Graph API / GraphQL when:**
 - Your data has complex relationships
 - You want to reduce the number of API calls
 - You need high flexibility in what you retrieve